

Asymmetry effects in untrusted noise security analysis

April 18, 2026

1 Imperfect gaussian measurements

We begin with a brief discussion of asymmetrical coherent measurements, schematically depicted in Fig. 1. Asymmetry effects are induced by imbalance (unequal transmission and reflection coefficients) of the beamsplitters and unequal quantum efficiency of the detectors. For detailed derivations of the expressions in this section see Ref. [1].

Q symbol of the POVM describing noisy homodyne measurements (see Fig. 1a) is given by

$$P_G^{(H)}(x) = \frac{1}{\sqrt{2\pi}\sigma_G} \exp \left[-\frac{(x - \langle \hat{x}_\phi \rangle)^2}{2\sigma_x} \right], \quad (1)$$

$$\langle \hat{x}_\phi \rangle \equiv \langle \alpha | \hat{x}_\phi | \alpha \rangle = 2 \operatorname{Re} \alpha e^{-i\phi}, \quad \phi = \arg \alpha_L, \quad (2)$$

$$\hat{x}_\phi = \hat{a} e^{-i\phi} + \hat{a}^\dagger e^{i\phi} \quad (3)$$

where α (α_L) is the complex amplitude of signal (LO) mode, $\hat{x}_\phi$ is the phase-rotated quadrature operator; x is the quadrature variable

$$x \equiv \frac{\mu}{(\eta_1 + \eta_2)CS |\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1 + \eta_2)CS} |\alpha_L| \quad (4)$$

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1 + \eta_2)CS]^2}. \quad (5)$$

Starting from the Q symbol (1) and passing through the characteristic function representation, we deduce the covariance matrix of the noisy homodyne measurements (see Ref. [1]):

$$\Sigma_m^{(H)} = \lim_{r \rightarrow +\infty} R(\phi) \operatorname{diag}(e^{-2r} + \sigma_N, e^{2r}) R^T(\phi), \quad (6)$$

$$\Sigma_{\text{id}}^{(H)} = \lim_{r \rightarrow +\infty} R(\phi) \operatorname{diag}(e^{-2r}, e^{2r}) R^T(\phi), \quad (7)$$

where $\Sigma_{\text{id}}^{(H)}$ is the covariance matrix of sharp measurement, $R(\phi) = \begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}$ is the rotation matrix; σ_N is the excess noise variance that accounts for asymmetry effects is given by

$$\sigma_N = \sigma_x - 1. \quad (8)$$

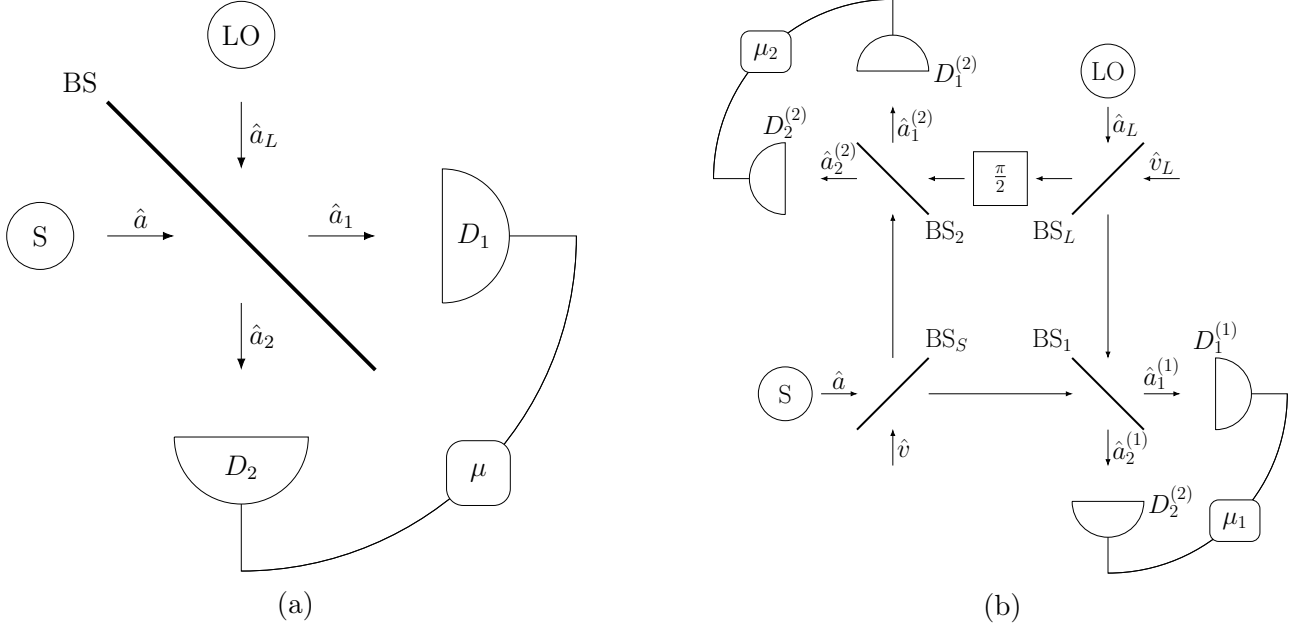


Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beamsplitter with transmission and reflection coefficients C and S , respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2 , and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double homodyne receiver. S (LO) is the source of signal (reference) mode, BS_S (BS_L) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS_i is the beam splitter of i th homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the i th homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of i th homodyne.

For double homodyne detection, Q symbol is given by

$$P_G^{(\text{DH})}(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}} \exp\left[-\frac{(x_1 - \text{Re } \alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2 - \text{Im } \alpha e^{-i\phi})^2}{\sigma_2}\right], \quad (9)$$

$$\sigma_G^{(i)} = (\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2) |\alpha_L^{(i)}|^2, \quad \alpha_L^{(1)} = C_L \alpha_L, \quad \alpha_L^{(2)} = -i S_L \alpha_L, \quad (10)$$

where x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)} + \eta_2^{(1)}) C_1 S_1 C_S} \left\{ \frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)} S_1^2 - \eta_2^{(1)} C_1^2) |\alpha_L^{(1)}| \right\}, \quad (11)$$

$$x_2 = \frac{1}{2(\eta_1^{(2)} + \eta_2^{(2)}) C_2 S_2 S_S} \left\{ \frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)} S_2^2 - \eta_2^{(2)} C_2^2) |\alpha_L^{(2)}| \right\}, \quad (12)$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \quad (13)$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)}) C_i S_i\right]^2} \quad (14)$$

give the quadrature variances σ_1 and σ_2 .

Analogously, we obtain the covariance matrix of the double homodyne POVM as:

$$\Sigma_m^{(\text{DH})} = R(\phi) \text{diag}(\delta_1, \delta_2) R^T(\phi), \quad (15)$$

$$\delta_i = 2\sigma_i - 1. \quad (16)$$

The POVM that describes sharp double homodyne measurements uses the set of squeezed states,

$$|z, r, \phi\rangle = \hat{R}(\phi) \hat{D}(z) \hat{S}(r) |0\rangle = \hat{D}(ze^{i\phi}) \hat{S}(re^{2i\phi}) |0\rangle, \quad (17)$$

where z is the coherent amplitude, r is real-valued squeezing parameter, and ϕ is squeezing phase; $\hat{D}(\beta)$ ($\hat{S}(\zeta)$) is the displacement (squeezing) operator given by

$$\hat{D}(\beta) = e^{\beta \hat{a}^\dagger - \beta^* \hat{a}}, \quad (18)$$

$$\hat{S}(\zeta) = e^{\frac{1}{2}(\zeta \hat{a}^{\dagger 2} - \zeta^* \hat{a}^2)}, \quad (19)$$

The covariance matrix of squeezed states reads

$$\Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^T(\phi), \quad (20)$$

and it is explicitly dependent on squeezing parameter r that lies in the interval defined by measurement asymmetry (see Ref. [1]):

$$r \in [r_2, r_1], \quad r_1 = \ln \delta_1^{1/2}, \quad r_2 = \ln \delta_2^{-1/2}. \quad (21)$$

2 Gaussian channels modeling imperfect measurements

Describing the measurement statistics of an arbitrary Gaussian state $\hat{\rho}_G$ with imperfect Gaussian measurements described by a noisy POVM $\hat{\Pi}_G$ amounts to computing the probability density of the outcome x , given by

$$p(x|\hat{\rho}_G) = \text{Tr} [\hat{\rho}_G \hat{\Pi}_G^{(x)}] = \text{Tr} [\mathcal{E}(\hat{\rho}_G) \hat{\Pi}_{\text{id}}^{(x)}] = \text{Tr} [\hat{\rho}_G \mathcal{E}^*(\Pi_{\text{id}}^{(x)})], \quad (22)$$

where $\{\Pi_{\text{id},x}\}$ is the POVM of the corresponding ideal (perfect) Gaussian measurement, and $\hat{\Pi}_G^{(x)} = \mathcal{E}^*(\Pi_{\text{id}}^{(x)})$ is the effective imperfect POVM describing the noisy measurement.

An important property is that the same measurement statistics (i.e., the same probability density $p(x|\hat{\rho}_G)$ for all Gaussian states $\hat{\rho}_G$) can be described by different Gaussian channels $\mathcal{E}$, provided their dual maps $\mathcal{E}^*$ yield the same effective POVM $\hat{\Pi}_G$. Moreover, as $\hat{\rho}_G$, $\hat{\Pi}_G$, and $\mathcal{E}$ are Gaussian, everything is characterized by the change of first and second moments.

In what follows, we will consider two distinct quantum channels to describe the imperfect measurements, namely attenuator channel and additive channel.

First, we shall use the attenuator quantum channel [2], characterized by matrices

$$X^{(\gamma)} = \cos \theta_\gamma \mathbb{1}, \quad Y^{(\gamma)} = \sin^2 \theta_\gamma \Sigma_{\text{aux}}^{(\gamma)}. \quad (23)$$

The action of the channel $\mathcal{E}_N^{(\gamma)}$ is defined as

$$\mathcal{E}_N^{(\gamma)}(\hat{\rho}) = \text{Tr}_{\text{aux}} \left[\hat{U}_\theta^{(\gamma)} (\hat{\rho} \otimes \hat{\rho}_{\text{aux}}^{(\gamma)}) \hat{U}_\theta^{(\gamma)\dagger} \right], \quad (24)$$

where $\hat{\rho}_{\text{aux}}^{(\gamma)}$ is a zero-mean Gaussian state with covariance matrix $\Sigma_{\text{aux}}^{(\gamma)}$, and $\hat{U}_\theta^{(\gamma)} = \exp \left[\theta_\gamma (a^\dagger \hat{b} - a \hat{b}^\dagger) \right]$ is the beam-splitter unitary operator, with $\hat{a}$ and $\hat{b}$ as the annihilation operators of the system and auxiliary modes, respectively.

Dual of (24), $\mathcal{E}_N^{*(\gamma)}$, transforms the POVM of the ideal measurement into the POVM of the noisy measurements by changing respective covariance matrix $\Sigma_{\text{id}}^{(\gamma)}$ into $\Sigma_m^{(\gamma)}$ as follows:

$$\mathcal{E}_N^{*(\gamma)} : \quad \Sigma_{\text{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2 \theta_\gamma} \Sigma_{\text{id}}^{(\gamma)} + \tan^2 \theta_\gamma \Sigma_{\text{aux}}^{(\gamma)}, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (25)$$

Now we derive explicit expressions coefficients of (25) (see Eq. (23)).

First, we will consider homodyne detection. It will be useful to put noisy POVM covariance matrices in the following form (see Eq. (6) and (7)):

$$\Sigma_m^{(\text{H})} = \Sigma_{\text{id}}^{(\text{H})} + \Sigma_N^{(\text{H})}, \quad (26)$$

$$\Sigma_N^{(\text{H})} = R(\phi) \text{diag}(\sigma_N, 0) R^T(\phi), \quad (27)$$

where we explicitly separated excess noise covariance matrix $\Sigma_N^{(\text{H})}$ and covariance matrix of sharp measurements $\Sigma_{\text{id}}^{(\text{H})}$.

Since covariance matrices of ideal and noisy measurements (see Eq. (7)) are taken in limit, implying that

$$\frac{1}{\cos^2 \theta} \Sigma_{\text{id}}^{(\text{H})} \sim \Sigma_{\text{id}}^{(\text{H})}, \quad e^{2r} + \text{const} \sim e^{2r}, \quad (28)$$

we may write the effective excess noise covariance matrix

$$\Sigma_N^{(\text{H}) \text{eff}} = R(\phi) \text{diag}(\sigma_N, \sigma_N) R^T(\phi), \quad (29)$$

so that environment is described by vacuum, i.e.

$$\Sigma_{\text{aux}}^{(\text{H})} = \mathbb{1}. \quad (30)$$

Solving for θ_{H} yields

$$\cos \theta_{\text{H}} = \sqrt{\frac{1}{\sigma_x}}, \quad \sin \theta_{\text{H}} = \sqrt{\frac{\sigma_N}{\sigma_x}}. \quad (31)$$

For double homodyne detection, first, we will separate the covariance matrices of sharp measurement and excess noise. From Eq. (15) and (20), we have

$$\Sigma_m^{(\text{DH})} = \Sigma_{\text{id}}^{(\text{DH})} + \Sigma_N^{(\text{DH})}, \quad (32)$$

where excess noise covariance matrix is given by

$$\Sigma_N^{(\text{DH})}(r) = \Sigma_m^{(\text{DH})} - \Sigma_{\text{id}}^{(\text{DH})} = R(\phi) \begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^T(\phi) \geq 0, \quad (33)$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \quad (34)$$

Note that the elements of $\Sigma_N^{(\text{DH})}$ are dependent on squeezing parameter that lies in the interval (21).

Now, we note that, since $\Sigma_m^{(\text{DH})}$ is anisotropic, as generally quadrature variances $\delta_{1,2}$ are not equal, $\delta_1 \neq \delta_2$, ancilla state must be anisotropic as well. Without loss of generality, we will consider the ancilla to be in pure state with $\det \Sigma_{\text{aux}}^{(\text{DH})} = 1$,

$$\Sigma_{\text{aux}}^{(\text{DH})} = R(\phi) \text{diag} \left(a, \frac{1}{a} \right) R^T(\phi), \quad a > 0. \quad (35)$$

From Eq. (25), we arrive at the expression for $\cos^2 \theta_{\text{DH}}$ in the form

$$\cos^2 \theta_{\text{DH}}(r) = \frac{e^{2r} + a(r)}{\delta_1 + a(r)}, \quad (36)$$

$$a(r) = \frac{1}{2} \left[-b + \sqrt{b^2 - 4c} \right], \quad b = \frac{-\delta_1 e^{-2r} + \delta_2 e^{2r}}{\sigma_N^{(2)}}, \quad c = \frac{-\sigma_N^{(1)}}{\sigma_N^{(2)}}. \quad (37)$$

It's worth noting that beyond the r interval ancilla becomes strictly unphysical, i.e. $a \in (0, +\infty) \iff r \in (r_2, r_1)$. This is due to $\sigma_N^{(i)}$ becoming negative, so that the argument of the square root in Eq. (37) becomes negative as well.

We will also consider classical mixing (additive noise) channel [2] used in Ref. [1]:

$$\Phi_N : \hat{\Pi}_{\text{id}} \mapsto \hat{\Pi}_G = \Phi_N(\hat{\Pi}_{\text{id}}) = \frac{2}{\pi \sqrt{\det \Sigma_N}} \int_{\mathbb{C}} d^2 \beta \exp\{-2\mathbf{s}^T \Sigma_N^{-1} \mathbf{s}\} \hat{D}(\beta) \hat{\Pi}_{\text{id}} \hat{D}(-\beta), \quad (38)$$

$$\mathbf{s}^T = (\text{Re}(\beta), \text{Im}(\beta)) = (\beta_1, \beta_2), \quad d^2 \beta = d\beta_1 d\beta_2. \quad (39)$$

This is a self-dual channel, $\Phi_N^* = \Phi_N$, that transforms the POVM of the ideal projective Gaussian measurements, $\hat{\Pi}_{\text{id}}$, into the POVM of the noisy measurements, $\hat{\Pi}_G$, by changing the covariance matrix Σ_{id} as follows

$$\Phi_N : \Sigma_{\text{id}} \mapsto \Sigma_G = \Sigma_{\text{id}} + \Sigma_N. \quad (40)$$

3 GMCS protocol

In the GMCS protocol, Alice sends coherent states $|\alpha = \alpha_1 + i\alpha_2\rangle$ drawn from the Gaussian prior

$$p_A(\alpha) = \frac{1}{\pi V_A} \exp \left[-\frac{|\alpha|^2}{2V_A} \right]. \quad (41)$$

The states pass the composite Gaussian channel $\mathcal{N}_{A \rightarrow B} = \Phi_{\xi_{\text{ch}}} \circ \mathcal{E}_{T_{\text{ch}}}$, where $\mathcal{E}_{T_{\text{ch}}}$ is pure loss channel with transmittance T_{ch} and Φ_{ξ} adds excess noise with covariance $\Sigma_{\xi_{\text{ch}}} = \xi_{\text{ch}} \mathbb{1}$. Then, the receiver uses either homodyne or double homodyne detection to measure the states. Mutual information between legitimate parties in each case is then given by the expressions (see Ref. [1] for derivation):

$$I_{\text{AB}}^{(\text{H})} = \frac{1}{2} \log \left[1 + \frac{4T_{\text{ch}} V_A}{\sigma_x + \xi_{\text{ch}}} \right], \quad (42)$$

$$I_{\text{AB}}^{(\text{DH})} = \frac{1}{2} \sum_i \log \left[1 + \frac{4T_{\text{ch}} V_A}{2\sigma_i + \xi_{\text{ch}}} \right], \quad (43)$$

where σ_x and σ_i given by Eqs. (5) and (14), respectively.

The covariance matrix of TMSV state with the Bob's mode transmitted through the noisy channel (see Ref. [3])

$$\mathcal{I}_A \otimes \Phi_{\xi_{\text{ch}}} \circ \mathcal{E}_{T_{\text{ch}}} : \quad \Sigma_{\text{TMSV}} = \begin{pmatrix} V\mathbb{1} & \sqrt{V^2 - 1}\sigma_z \\ \sqrt{V^2 - 1}\sigma_z & V\mathbb{1} \end{pmatrix} \mapsto \Sigma_{\text{AB}} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_B\mathbb{1} \end{pmatrix}, \quad (44)$$

where $\mathcal{I}_A$ is the identity channel acting on the subsystem A and the parameters of the covariance matrix Σ_{AB} are given by

$$V = 1 + 4V_A, \quad c = \sqrt{T_{\text{ch}}(V^2 - 1)}, \quad (45)$$

$$V_B = T_{\text{ch}}(V - 1) + 1 + \xi_{\text{ch}}. \quad (46)$$

In untrusted noise scenario, where excess noise caused by asymmetry effects is assumed to be in control of Eve, channel that models the excess noise can be modeled in two equivalent ways, per Eq. (24) and Eq. (38).

We consider local action of either attenuator channel $\mathcal{E}_N^{(\gamma)}$ (24) or additive noise channel $\Phi_N^{(\gamma)}$ (38), on second mode of TMSVS shared by Alice and Bob (44). In both cases, mutual information is given by $I_{\text{AB}}^{(\gamma)}$ (42), (43), as it is classical.

It follows

$$\mathcal{I}_A \otimes \mathcal{E}_N^{(\gamma)} : \Sigma_{\text{AB}} \mapsto \Sigma_{\text{AB}}^{(\gamma; \mathcal{E}_N^{(\gamma)})} = \begin{pmatrix} V\mathbb{1} & c \cos \theta_\gamma \sigma_z \\ c \cos \theta_\gamma \sigma_z & \cos^2 \theta_\gamma \left(V_B + \tan^2 \theta_\gamma \Sigma_{\text{aux}}^{(\gamma)} \right) \end{pmatrix}, \quad (47)$$

$$\mathcal{I}_A \otimes \Phi_N^{(\gamma)} : \Sigma_{\text{AB}} \mapsto \Sigma_{\text{AB}}^{(\gamma; \Phi_N^{(\gamma)})} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_B\mathbb{1} + \Sigma_N^{(\gamma)} \end{pmatrix}, \quad \gamma \in \{\text{H}, \text{DH}\}. \quad (48)$$

When the channel $C^{(\gamma)} \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}$ is under control of the eavesdropper, Eve holds a purification of the state $\hat{\rho}_{\text{AB}}^{(\gamma; C^{(\gamma)})} = \mathcal{I}_A \otimes C^{(\gamma)}(\hat{\rho}_{\text{AB}})$ and the measurements performed by the receiver are assumed to be noiseless. In this scenario, the difference between the joint and conditional entropies

$$\chi_{\text{EB}}^{(\gamma; C^{(\gamma)})} = S_{\text{E}} - S_{\text{E|B}} = S_{\text{AB}}^{(\gamma; C^{(\gamma)})} - S_{\text{A|B}}^{(\gamma; C^{(\gamma)})}, \quad C^{(\gamma)} \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}. \quad (49)$$

gives the Holevo information. The joint entropy

$$S_{\text{AB}}^{(\gamma; C^{(\gamma)})} = g(\nu_1^{(\gamma; C^{(\gamma)})}) + g(\nu_2^{(\gamma; C^{(\gamma)})}), \quad (50)$$

$$g(\nu) = \frac{\nu + 1}{2} \log \frac{\nu + 1}{2} - \frac{\nu - 1}{2} \log \frac{\nu - 1}{2}$$

then can be expressed in terms of the symplectic eigenvalues of the covariance matrix (48) given by

$$\nu_{1,2}^{(\gamma; C^{(\gamma)})} = \sqrt{\Delta_{\gamma; C^{(\gamma)}}/2 \pm \sqrt{\Delta_{\gamma; C^{(\gamma)}}^2/4 - D_{\gamma; C^{(\gamma)}}}}, \quad D_{\gamma; C^{(\gamma)}} = \det \Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}, \quad (51)$$

$$\Delta_{\gamma; C^{(\gamma)}} = \det([\Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}]_{11}) + \det([\Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}]_{22}) + 2 \det([\Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}]_{12}), \quad (52)$$

where $[\Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}]_{ii}$ ($[\Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{\text{AB}}^{(\gamma; C^{(\gamma)})}$.

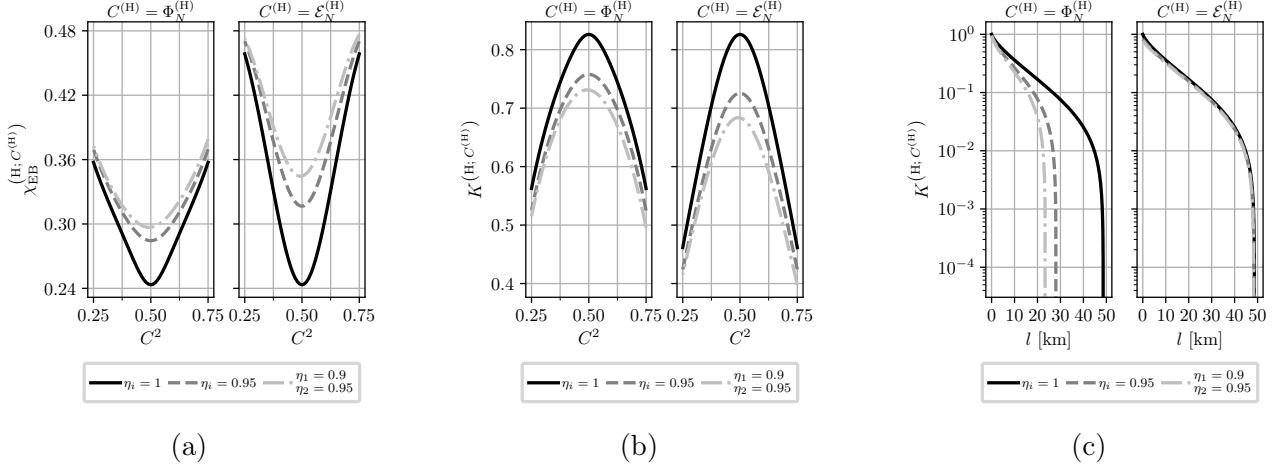


Figure 2: (a) Holevo information, $\chi_{EB}^{(H;C)}$ and (b) asymptotic secret fraction, $K^{(H;C)}$, as a function of the beam splitter transmission, C^2 , of the homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5$; $T_{ch} = 0.95$; $\xi_{ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(H)}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{ch} = 10^{-2}$; losses are 20 dB per 100 km.

Conditional entropies $S_{E|B} = S_{A|B}^{(\gamma)}$ are calculated from respective conditional covariance matrices,

$$S_{A|B}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det \Sigma_{A|B}^{(\gamma)}}, \quad (53)$$

$$\Sigma_{A|B}^{(\gamma)} = V \mathbb{1} - c^2 \sigma_z (V_B \mathbb{1} + \Sigma_m^{(\gamma)})^{-1} \sigma_z. \quad (54)$$

Direct computations show that these conditional matrices and therefore their symplectic eigenvalues are independent on channel model. Symplectic eigenvalues are given by:

$$\nu_3^{(H)} = \sqrt{V \left(V - \frac{c^2}{V_B + \sigma_N} \right)}, \quad (55)$$

$$\nu_3^{(DH)} = V \sqrt{\frac{(V_B + \delta_1 - \frac{c^2}{V})(V_B + \delta_2 - \frac{c^2}{V})}{(V_B + \delta_1)(V_B + \delta_2)}}, \quad (56)$$

From mutual information and Holevo information evaluation of the asymptotic secret fraction is done per Devetak-Winter formula [4]

$$K^{(\gamma; C^{(\gamma)})} = \beta I_{AB}^{(\gamma; C^{(\gamma)})} - \chi_{EB}^{(\gamma; C^{(\gamma)})}, \quad \gamma \in \{H, DH\}, C^{(\gamma)} \in \{\Phi_N^{(\gamma)}, \mathcal{E}_N^{(\gamma)}\}. \quad (57)$$

As symplectic eigenvalues for additive noise channel are calculated per formulas Eq. (51) and do not allow for simplification, in the following sections we focus only on analytic expressions for symplectic eigenvalues for attenuator channel model.

4 Homodyne detection

For attenuator channel, from Eq. (47) and (31) we have

$$\Sigma_{AB}^{(H; \mathcal{E}_N^{(H)})} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}(V_B + \sigma_N)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_B\mathbb{1} \end{pmatrix}, \quad (58)$$

$$\tilde{V}_B \equiv T(V - 1) + 1 + \xi, \quad (59)$$

$$\tilde{c} \equiv \sqrt{T(V^2 - 1)}, \quad (60)$$

where we defined scaled Bob's variance and scaled correlations between Alice and Bob $\tilde{V}_B$ and $\tilde{c}$, using total channel transmission T that consists of channel transmission T_{ch} and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ :

$$T = T_{\text{ch}} \cdot \frac{1}{\sigma_x}, \quad (61)$$

$$\xi \equiv \xi_{\text{ch}} \cdot \frac{1}{\sigma_x}. \quad (62)$$

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (61) and channel noise (62) by a factor of σ_x^{-1} .

Symplectic eigenvalues of $\Sigma_{AB}^{(H; \mathcal{E}_N^{(H)})}$, $\nu_{1,2}^{(H; \mathcal{E}_N^{(H)})}$, can be computed using formulas [5]:

$$\nu_{1,2}^{(H; \mathcal{E}_N^{(H)})} = \frac{z \pm [\tilde{V}_B - V]}{2}, \quad z = \sqrt{(V + \tilde{V}_B)^2 - 4\tilde{c}^2}, \quad (63)$$

and then substituted in Eq. (50) to calculate joint entropy $S_{AB}^{(H; \mathcal{E}_N^{(H)})}$.

Fig. 2 illustrates the effects of homodyne measurement asymmetry on Holevo information (49) and asymptotic secret fraction (57) versus beam splitter transmission C^2 , as well as secret fraction versus transmission length, for both additive noise (48) and attenuator (47) channels. While channel noise is reduced in this setup (62), so is channel transmission (61), and, since transmission is scaled by modulation variance $V \gg 1$ in every expression, as well as reduced mutual information with greater asymmetry per Eq. (42), attenuator channel has negative effect. As shown in Fig. 2a and 2b, the attenuator channel results in higher Holevo information, and thus lower secret fraction, at high transmission for fixed asymmetry parameters. However, Fig. 2c reveals that the attenuator channel achieves significantly greater reach (maximum transmission length at which a positive key rate is obtained) than the additive noise channel.

5 Double homodyne

For attenuator, from Eq. (47) and (36) we have

$$\Sigma_{AB}^{(\text{DH}; \mathcal{E}_N^{(\text{DH})})} = \begin{pmatrix} V\mathbb{1} & \cos^2 \theta^{(\text{DH}; \mathcal{E}_N^{(\text{DH})})} \left(V_B + \delta_1 - \frac{e^{2r}}{\cos^2 \theta^{(\text{DH}; \mathcal{E}_N^{(\text{DH})})}} \right) & 0 \\ c \cos \theta^{(\text{DH}; \mathcal{E}_N^{(\text{DH})})} \sigma_z & 0 & \cos^2 \theta^{(\text{DH}; \mathcal{E}_N^{(\text{DH})})} \left(V_B + \delta_2 - \frac{e^{-2r}}{\cos^2 \theta^{(\text{DH}; \mathcal{E}_N^{(\text{DH})})}} \right) \end{pmatrix}. \quad (64)$$

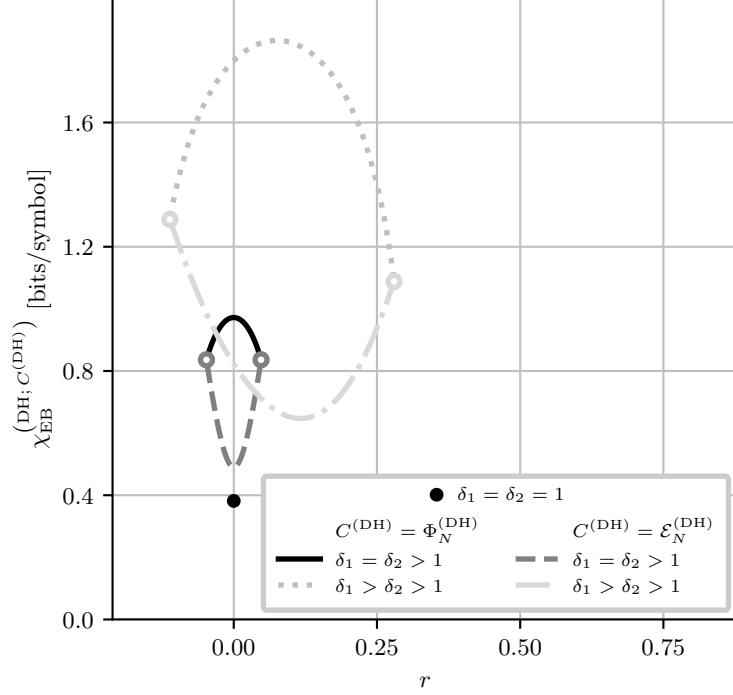


Figure 3: Holevo information, $\chi_{EB}^{(DH; C)}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. Black dot corresponds to the case of perfect measurements. Solid and dashdotted curves correspond to additive noise channel model, while dashed and dotted curves are for attenuator channel model. Color-matched white-filled dots mark points where Holevo information is undefined at the boundaries of the r interval (21) (see Eq. (37) and main text). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dashdotted and dotted curves. The channel parameters are: $V = 5$; $T_{ch} = 0.95$; and $\xi_{ch} = 10^{-2}$.

Symplectic eigenvalues of this matrix can be calculated using invariants similarly to Eq. (51). It is obvious that these symplectic eigenvalues, and thereby joint entropy, is explicitly depended on squeezing parameter r (21), like in the case of additive noise channel.

Fig. 3 illustrates dependence of Holevo information on the squeezing parameter, for both additive noise and attenuator channels. Referring to Fig. 3, in the limiting case of ideal double homodyne with $\delta_1 = \delta_2 = 1$, only one value of $r = 0$ is allowed (see Ref. [1]), with equal values of Holevo information for both channels (black dot on plot).

When $\delta_1 = \delta_2 > 1$, dependence of Holevo information on squeezing parameter is symmetrical around $r = 0$ for both channels. For additive noise channel, $r = 0$ is the point of maximal Holevo information, while for attenuator channel $r = 0$ corresponds to minimal Holevo information. Moreover, while Holevo information is undefined at $r = r_{2,1}$ (see Eq. (36)), the values tend to the same values of joint entropy in the case of additive noise channel, as signified by color-matched white-filled dots in Fig. 3.

In asymmetrical case where $\delta_1 \neq \delta_2$, Holevo information for both channel models become asymmetrical relative to their extrema, resulting in unequal joint entropies at $r = r_{2,1}$. For additive noise channel, Holevo information has a distinctive maxima, while for attenuator channel Holevo information has a distinctive minima.

Fig. 4 illustrates lower and upper bounds of the asymptotic secret fraction versus transmission length, for both additive noise (48) and attenuator (47) channels. Note that for chosen $\sigma_x^{(i)}$ lower bounds for both channels coincide, because for $\sigma_x^{(i)} \approx 1$ maxima of joint entropy for

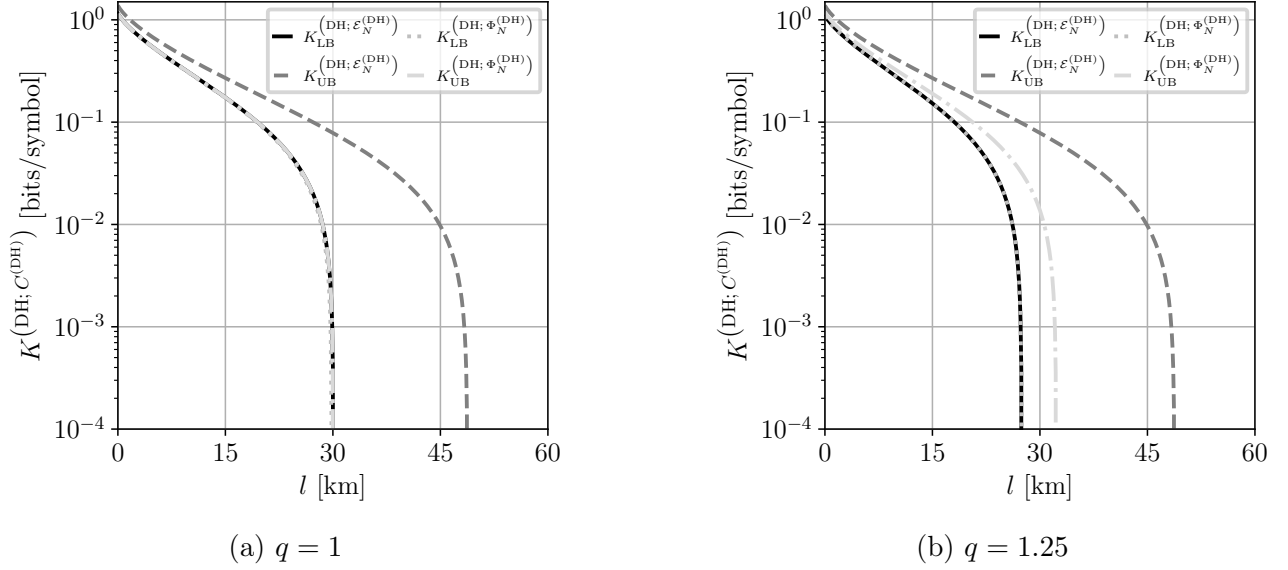


Figure 4: Asymptotic secret fraction $K^{(\text{DH};C)}$ versus channel length l , computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq.(14)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for additive noise and attenuator channel models (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\text{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

both models is either at $r = r_1$ or $r = r_2$. Moreover, the upper bound for additive channel in Fig. 4a is very close (0.2 km difference) to the lower bound.

As seen from Fig. 4, difference between lower and upper bound of the asymptotic secret fraction is greater in

Fig. 5 illustrates the dependence of Holevo information (49) and asymptotic secret fraction (57) on the signal beam splitter transmission C_S^2 , as well as secret fraction versus transmission length if the receiver uses double homodyne detection, for both additive noise (48) and attenuator (47) channels. Unlike the homodyne detection case (see Fig. 2), Holevo information is significantly lower for attenuator channel, except for the ideal measurements case ($\eta_{1,2}^{(1,2)} = 1$ and whole C_S^2 axis, unlike in homodyne case where sharp measurements are strictly $\eta_{1,2} = 1$ and $C^2 = 0.5$) as both channels have no effect and thereby have equal joint entropies (black curve). High difference in Holevo information is due to the dependence of joint entropy on r (see Fig. 3) and fixing $r = \frac{r_2+r_1}{2}$ for both curves, effectively minimizing Holevo information for attenuator channel and maximizing it for additive noise channel.

Analogously to homodyne detection case (see Fig. 2c), in double homodyne detection case the attenuator channel has greater reach than the additive noise channel, as illustrated by Fig. 5c.

References

- [1] A. S. Naumchik, Roman K. Goncharov, and Alexei D. Kiselev. *Asymmetry effects in homodyne and heterodyne measurements: Positive operator-valued measures and asymptotic security of Gaussian continuous variable quantum key distribution*. 2025. arXiv: [2512.22591](https://arxiv.org/abs/2512.22591) [quant-ph]. URL: <https://arxiv.org/abs/2512.22591>.
- [2] Alessio Serafini. *Quantum Continuous Variables: A Primer of Theoretical Methods*. 2nd. Boca Raton: CRC Press, Taylor & Francis Group, 2023, p. 362. DOI: [10.1201/9781003250975](https://doi.org/10.1201/9781003250975).

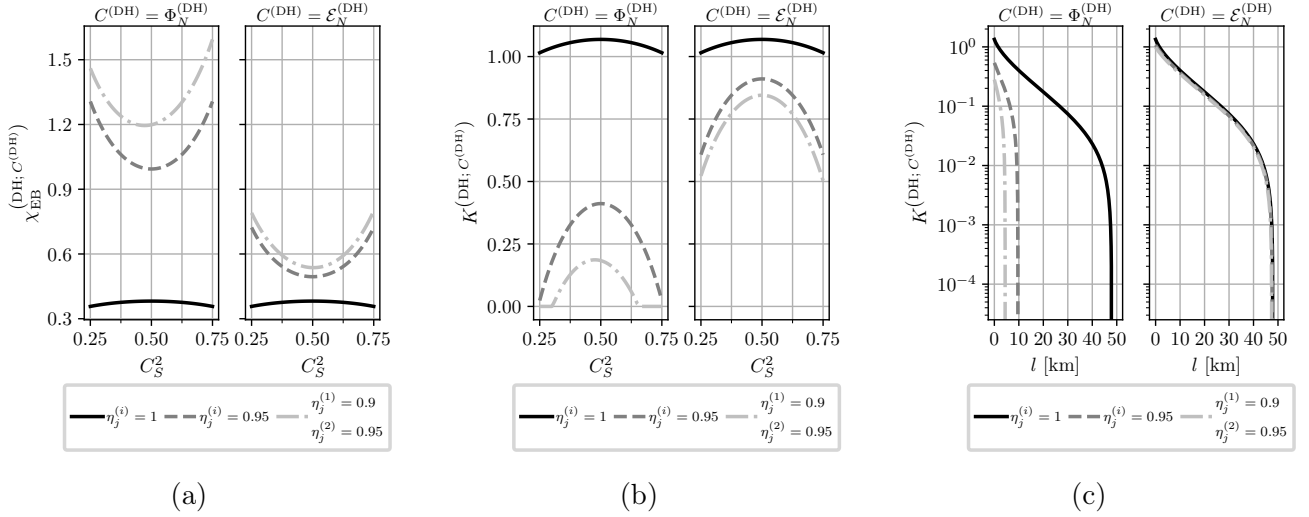


Figure 5: (a) Holevo information, $\chi_{\text{EB}}^{(\text{DH}; C)}$ and (b) asymptotic secret fraction, $K^{(\text{DH}; C)}$, as a function of the signal beam splitter transmission, C_S^2 , of the double homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5$; $T_{\text{ch}} = 0.95$; $\xi_{\text{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(\text{DH}; C)}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\text{ch}} = 10^{-2}$; losses are 20 dB per 100 km.. For both channel models $r = \frac{r_2 + r_1}{2}$ (see main text).

- [3] Fabian Laudenbach et al. “Continuous-Variable Quantum Key Distribution with Gaussian Modulation—The Theory of Practical Implementations”. In: *Advanced Quantum Technologies* 1.1 (2018), p. 1800011. DOI: [10.1002/qute.201800011](https://doi.org/10.1002/qute.201800011).
- [4] Igor Devetak and Andreas Winter. “Distillation of secret key and entanglement from quantum states”. In: *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 461.2053 (2005), pp. 207–235. DOI: [10.1098/rspa.2004.1372](https://doi.org/10.1098/rspa.2004.1372).
- [5] Christian Weedbrook et al. “Gaussian quantum information”. In: *Rev. Mod. Phys.* 84 (2 May 2012), pp. 621–669. DOI: [10.1103/RevModPhys.84.621](https://doi.org/10.1103/RevModPhys.84.621).